

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Cryptography, Privacy Enhancing Techniques

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++, C, C#

Machine Learning: Machine Learning, Deep Learning, Supervised Learning, Unsupervised Learning, Predictive Modeling, Model Training, Model Evaluation, Model Validation, Feature Engineering, Hyperparameter Tuning, Neural Networks

Deep Learning: PyTorch, NumPy, Pandas, Scikit-learn, CNNs, RNNs, LSTMs, Res-UNet, Transfer Learning, Image Synthesis

Computer Vision: Computer Vision, Image Processing, Object Detection, Object Tracking, Pose Estimation, Multi-Object Tracking, Image Segmentation, Image Alignment, Inpainting, Keypoint Detection

Generative AI & LLMs: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), RAG Pipelines, LLM Integration, Vector Embeddings, Semantic Search, Prompt Engineering, Token Optimization, Model Context Protocol (MCP), LangChain, FAISS, Ollama

AI Model Evaluation: Model Evaluation, Performance Benchmarking, PSNR, SSIM, FID, LPIPS, mAP, MOTA, IDF1, Accuracy, Confusion Matrix

AI Tools & Platforms: Streamlit, Docker, Git, Jupyter, Linux

PROJECTS

Local LLM-Powered News Research Platform

Python, LangChain, Streamlit, FAISS, Ollama

- Developed an end-to-end local ETL pipeline using Streamlit and WebBaseLoader to ingest, clean, and process unstructured web content on-demand
- Designed a local vector storage architecture using FAISS and nomic-embed-text to serialize high-dimensional embeddings, bypassing redundant ingestion steps and reducing processing latency
- Integrated a local LLM retrieval system via LangChain to orchestrate semantic queries against the vectorized data using local Llama 3.2 instances

Pacman Search Algorithms

Python | Data Structures & Algorithms | BFS | A* | Heuristics

- Designed and implemented search algorithms in Python, defining custom problem states, successor functions, goal conditions, and state representations for maze-based environments
- Implemented Breadth-First Search (BFS) and A* search with admissible heuristics to efficiently navigate complex mazes while minimizing path cost
- Developed a greedy approximation strategy for large state spaces where optimal search was computationally expensive, improving scalability for complex search problems

PROFESSIONAL EXPERIENCE

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

Project: [MiGa: Multi-Chicken Gait Assessment](#)

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking (MOT), and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a Multi-Bird Detection split of 4,569 frames with 11,422 bounding boxes and a Gait-Score Regression split spanning 55 expert-annotated sequences
- Implemented spatial normalization to map keypoints into local coordinate spaces to eliminate translational variance, achieving a keypoint mAP of 91.83% at 1,250 FPS alongside 0.93 MOTA and 0.91 IDF1 tracking accuracy

Project: [DualFit: Two-Stage Virtual Try-On via Warping and Synthesis](#)

- Developed a PyTorch-based two-stage Virtual Try-On (VTON) model combining flow-based garment warping with Res-UNet image synthesis, with preprocessing and inpainting pipelines to improve garment alignment and visual realism
- Trained and evaluated the model on the VITON-HD dataset, achieving 24.9 PSNR, 0.913 SSIM, and 5.261 FID

Project Engineer, Soliton Technologies, India

July 2022 – June 2024

- Developed a C# WPF application integrating Python and LabVIEW-based test workflows to automate testing and improve execution efficiency.
- Served as the sole developer for a customized Medtronic solution, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements.
- Implemented reusable automation and validation workflows, collaborating with cross-functional teams to optimize processes, improve test coverage, and troubleshoot system-level issues.